

Engineering Mathematics II

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\frac{dy}{dx} = f'(x)$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\int_a^b f(x) dx$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\vec{R} = \vec{A} + \vec{B}$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$AX = B$$

Section B

5. Differentiate a cost function. **(10 marks)**
6. Calculate resultant forces in a simple frame. **(10 marks)**
7. Use matrices to solve building quantity equations. **(10 marks)**
8. Create one realistic building technology problem based on this topic and solve it fully. **(20 marks)**